HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented new features for building and structural engineering software, focusing on **Digital Twin Enablement** and enhancing product maintainability.
- Collaborated within a small, multi-disciplinary team to develop, test, and document software solutions, adhering to **Agile and Scrum** methodologies.
- Investigated and resolved complex technical issues, proposing software designs with **optimal patterns** and considering broader system impacts.
- Contributed to code reviews, ensuring adherence to quality standards and providing constructive feedback on **maintainable, testable, and secure code**.
- Researched and evaluated new technologies to support product evolution and maintainability for enterprise applications.
- Developed comprehensive unit and integration tests to validate functionality and ensure adherence to use cases and quality criteria.
- Containerized software components using **Docker** for consistent deployment across development and production environments on **AWS**.

SKILLS

Languages – Python, C++, C#, .NET

AI Tools – GitHub Copilot, Claude, Cursor

Development Methodologies – Agile, Scrum, SDLC, SOLID, DRY principles, Object-Oriented Programming

Tools & Platforms – Git, Azure DevOps, Jira, TFS, ADO

Testing – Unit testing frameworks, Integration testing, System testing

Domain – Building Engineering, Structural Engineering, Infrastructure Design, Geometry, Mathematics, Trigonometry

PROJECTS

Digital Twin Integration for Building Design Software

- Developed core **Python** modules to process and integrate geospatial data, forming the foundation for Digital Twin capabilities.
- Implemented RESTful APIs using **Flask** to expose processed engineering data for consumption by downstream visualization services.
- Authored unit tests with **pytest**, achieving over 85% code coverage for critical data processing logic.

Cloud-Native Refactoring of Engineering Simulation Tools

- Refactored legacy C++ simulation components into modular **.NET** services, improving maintainability and testability.
- Designed and implemented a **REST API** layer for simulation services, enabling integration with cloud-based project management platforms.
- Containerized the refactored services using **Docker** and deployed them to an **Azure Kubernetes Service (AKS)** cluster.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)